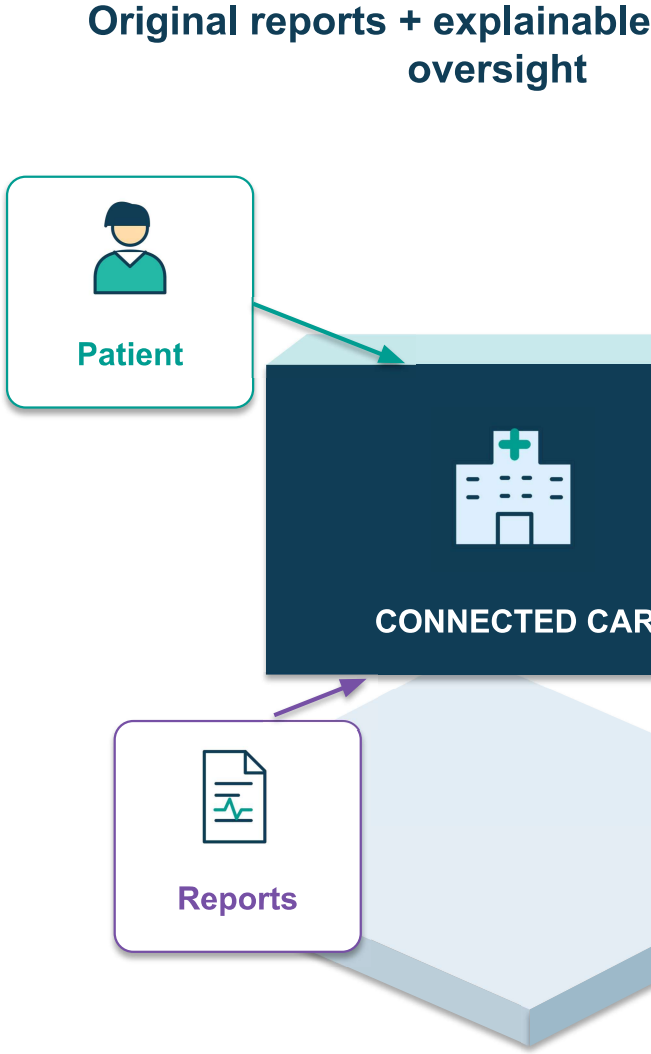
# MediConnect AI

## AI-powered medical report understanding and connected patient care

| | |
|---|---|
| **Problem Statement ID** | T01 |
| **Problem Statement Title** | AI-assisted medical reports & patient care |
| **Theme** | Healthcare / MedTech — proposed |
| **PS Category** | Software |
| **Team ID** | AAYU-WF9D |
| **Team Name** | S Square |

**UNDERSTAND • ORGANIZE • CONNECT**



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# MEDICONNECT AI

Your report becomes understandable, retrievable and ready for authorized clinical review.

## PROPOSED SOLUTION

**One healthcare platform for report explanations, secure medical history and doctor-led follow-up.**

### Reports patients can understand

OCR + structured extraction turn PDF / image reports into simple, source-grounded explanations.

### History doctors can review

A dated timeline brings original reports, compatible test trends and reviewed summaries together.

### Care workflows in one place

Appointments, follow-ups and itemized bill checks connect the report to the next care step.

## WHAT WE WILL BUILD

### PATIENT MODULE

Secure upload, plain-language explanations, med

### DOCTOR MODULE

Authorized records, source-linked summaries, re
prescriptions.

### HOSPITAL MODULE

Profiles, staff, schedules and role-based administ

### BILLING MODULE

Flag possible duplicate charges and arithmetic er
clarification.

## WHY WE STAND OUT

### Source-linked AI

Original evidence stays visible.

### Rules before reasoning

Values and units are validated.

### Doctor in control

Approve, correct or reject.

### Cor

Rec

# TECHNICAL APPROACH

Modular services + validated extraction + privacy-by-design + mandatory clinician oversight.

## ZONE 1 • USERS & PORTALS

- Patient portal
- Doctor portal
- Hospital admin

React + Tailwind
HTTPS / role-based access

## ZONE 2 • APPLICATION LAYER

**API GATEWAY**
Authentication • consent • permissions

**REPORT SERVICE**
Upload • records • summary versions

**CARE SERVICE**
Review • appointments • notes • Rx

**BILL REVIEW**
Duplicate / arithmetic discrepancy flags

Python + FastAPI • Docker

## ZONE 3 • AI + VALIDATION

**TEXT EXTRACTION**
Digital PDF text / OCR

**VALIDATE DATA**
Values, units, ranges, dates

**GROUNDED SUMMARY**
LLM uses verified source data

PaddleOCR / Tesseract + LLM

## ZONE 4

Profiles
appoint

Origina
Encrypt

## END-TO-END PROCESS

01 Consent + login → 02 Upload report → 03 OCR + extract → Data valid?

Yes → 04 Explain + timeline → 05 Doctor review → 0 f

No → flag issue, correct data or re-upload

NO AUTONOMOUS DIAGNOSIS

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# FEASIBILITY AND VIABILITY

A focused MVP first; advanced features follow demonstrated safety and usefulness.

## FEASIBILITY

### Buildable core stack

React, FastAPI, PostgreSQL and open-source OCR enable a modular first implementation.

### Complete minimum viable demo

Login → upload → grounded explanation → timeline → doctor review → booking.

### Testable, not assumed

Measure extraction, grounding, access control, retrieval time and booking correctness.

## CHALLENGES & STRATEGIES

### OCR / AI reliability

Confidence gates + source-linked findings + do

### Privacy & access

Consent, least privilege, encryption, audit and

### Clinical / integration readiness

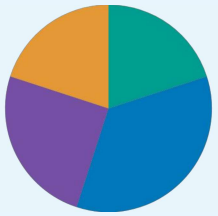
Validate lab units and ranges; require expert re integrations.

## VIABILITY & ROADMAP

**Modular rollout enables staged testing.**

| 01 | 02 | 03 | 04 |
|---|---|---|---|
| Roles + consent | Reports + AI | Review + booking | Validate / extend |

### Proposed development effort



20% Foundation

35% AI reports

Illustrative allocation — not measure

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# IMPACT AND BENEFITS

Expected value for patients, clinicians and hospitals — all improvements require pilot measurement.

## SOCIAL VALUE

Plain-language explanations and accessible records can support patient understanding.

## WORKFLOW VALUE

Organized history can reduce repeated record-search and consultation preparation work.

## ECONOMIC VA

Transparent bill re
clarify duplicate er
errors.

## STAKEHOLDERS & IMPACT

### 01 PATIENT

Upload once; understand and revisit reports.

### 02 DOCTOR

Review original findings and organized history.

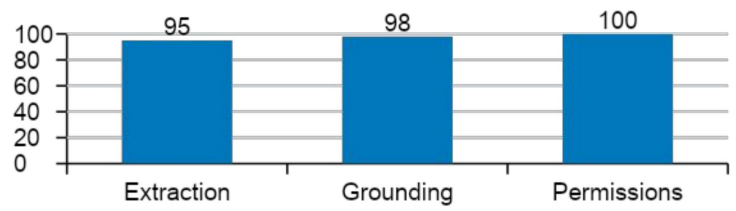
### 03 HOSPITAL

Coordinate records, schedules and bill queries.

### 04

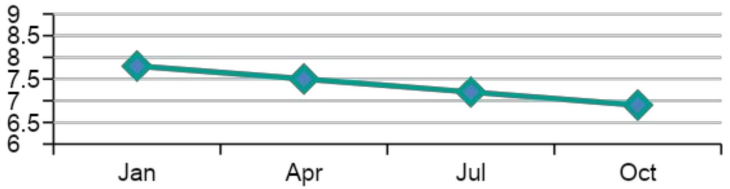
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## Proposed validation targets (%)




Proposed thresholds — no measured results.

## Medical trend demo: HbA1c (%)




Synthetic example — not real patient or efficacy data.

## OUR CO

**AI assists; cl
Consent and
Never delay e**

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# RESEARCH AND REFERENCES

Background sources guide the design; they do not establish approvals, partnerships or clinical efficacy

### WHO • Ethics and governance of AI for health

Human oversight, accountability, privacy and transparent use.

who.int/publications/i/item/9789240029200

### HL7 FHIR • Healthcare interoperability

Reference for future structured, authorized health-data exchange.

hl7.org/fhir/overview.html

### ABDM • Indian digital-health ecosystem

Official ecosystem context; integrations require permission.

abdm.gov.in

### Implementation documentation

Suggested stack: FastAPI, PostgreSQL and PaddleOCR.

fastapi.tiangolo.com • postgresql.org/docs • paddleocr.ai

**PROJECT**



**Scan to open the supp**

## EVIDENCE & TESTING PLAN

Synthetic / de-identified records • original-vs-extracted checks • summary grounding • access tests • booking + recovery

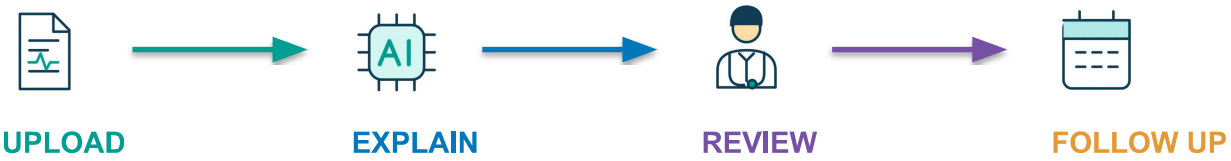
THANK YOU

# MediConnect AI

## Smarter reports.
## Clearer understanding.
## Connected healthcare.




**One upload → organized history → doctor-reviewed care**




AI assists healthcare professionals. It does not independently diagnose or prescribe.